CycleGAN: Horse to Zebra Unpaired Image-to-Image Translation

with Proposed Perceptual Cycle Consistency Loss

Deep Learning for Perception
Final Semester Project Report

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1. Task Definition

This project addresses the problem of unpaired image-to-image translation, a computer vision task where the goal is to learn a mapping between two visual domains without any paired training examples. Specifically, the task involves translating photographs of horses into photographs that resemble zebras, and vice versa.

The framework used to solve this task is CycleGAN, proposed by Zhu et al. (2017). CycleGAN learns two translation functions simultaneously: one that maps horse images to the zebra domain, and another that maps zebra images back to the horse domain. A cycle consistency constraint is enforced so that translating an image from one domain to the other and back again recovers the original image. This constraint allows the model to train without any paired correspondence between the two domains.

Beyond replicating the standard CycleGAN baseline, this project also proposes a modification to the cycle consistency loss. The standard approach penalises pixel-level differences between the original and reconstructed images, which can lead to blurry outputs. The proposed modification replaces this pixel-level penalty with a perceptual loss computed in the feature space of a pretrained VGG-16 network. The hypothesis is that comparing images at the level of learned visual features produces sharper, more structurally faithful reconstructions.

2. Dataset Description

The Horse2Zebra dataset, introduced alongside the CycleGAN paper (Zhu et al., 2017), is used for this project. The dataset consists of RGB colour photographs sourced from the ImageNet horse synset (n02381460) and zebra synset (n02391049). All images are in JPEG format.

The dataset is split into four directories. The table below summarises the composition of each split:

Directory	Domain	Number of Images
trainA	Horses (training)	1,067
trainB	Zebras (training)	1,334
testA	Horses (test)	120
testB	Zebras (test)	140

The dataset does not contain any labels in the conventional sense. There are no bounding boxes, segmentation masks, or class annotations. The only supervision available is domain membership, meaning each image is known to belong to either the horse domain or the zebra domain. The task is therefore fully unsupervised with respect to image content.

Importantly, the training images are unpaired. There is no correspondence between any horse image in trainA and any zebra image in trainB. The model must discover the visual differences between the two domains entirely from their respective distributions.

It is worth noting that the zebra training set is approximately 25 percent larger than the horse training set. This imbalance partially contributes to the observed asymmetry in translation quality, where the horse to zebra direction tends to produce better results than the reverse.

3. Data Pre-processing

3.1 Training Pipeline

Each training image undergoes the following sequence of transformations before being passed to the network:

- Resize: Each image is resized to 286 by 286 pixels using bicubic interpolation. This is intentionally larger than the final training resolution to allow for spatial jitter in the next step.
- Random Crop: A 256 by 256 patch is randomly cropped from the 286 by 286 image. This introduces positional augmentation and prevents the network from overfitting to edge locations.
- Random Horizontal Flip: The image is flipped horizontally with a probability of 0.5. This effectively doubles the number of training examples and improves the model's left-right invariance.
- Normalisation: Pixel values, originally in the range 0 to 1, are normalised to the range negative 1 to positive 1 using a mean of 0.5 and a standard deviation of 0.5 for each colour channel. This range matches the output range of the generator's Tanh activation.

3.2 Test and Inference Pipeline

During evaluation and inference, augmentation is disabled to ensure deterministic results. Each image is resized directly to 256 by 256 pixels using bicubic interpolation, followed by the same normalisation step as training. No random cropping or flipping is applied.

No additional pre-processing steps such as noise reduction, histogram equalisation, or colour jitter are applied at any stage. Since the domain difference between horses and zebras is primarily one of texture rather than illumination or geometry, aggressive colour augmentation would interfere with the style signal the model needs to learn.

The data loading class samples images from Domain A and Domain B independently within each batch. No attempt is made to create or enforce any pairing between horse and zebra images during training.

4. Network Architecture

The overall system consists of four neural networks trained simultaneously: two generators and two discriminators. The generators learn the translation mappings between domains, while the discriminators assess whether generated images are realistic.

4.1 Generator

The generator follows the ResNet-based encoder-decoder architecture proposed by Johnson et al. (2016), adapted for use in CycleGAN. The network consists of three stages: a downsampling encoder, a sequence of residual blocks that transform the feature representation, and an upsampling decoder that reconstructs the image at full resolution.

The encoder begins with a large-kernel convolutional layer that processes the full-resolution input and produces 64 feature maps. This is followed by two strided convolutional layers that progressively halve the spatial resolution while doubling the number of channels, arriving at a compact representation of 256 channels at one quarter of the original resolution.

The bottleneck consists of nine residual blocks, each containing two convolutional layers with a skip connection. This depth is recommended by Zhu et al. for images of 256 by 256 resolution. The residual connections allow gradient flow during training and help the generator preserve content structure while modifying style.

The decoder mirrors the encoder using two transposed convolutional layers to progressively restore spatial resolution, followed by a final convolutional layer that maps to three output channels. A Tanh activation at the output confines pixel values to the range negative 1 to positive 1.

Throughout the generator, instance normalisation is used in place of batch normalisation. Instance normalisation normalises each image independently, which is important for style-transfer tasks where per-image statistics are meaningful. Reflection padding is used throughout to reduce border artifacts that commonly arise with zero-padding. Each generator contains approximately 11.4 million trainable parameters.

4.2 Discriminator

The discriminator uses the PatchGAN architecture (Isola et al., 2017), which classifies overlapping local patches of the input image as real or fake rather than classifying the entire image with a single scalar output. This design encourages the generator to produce realistic high-frequency textures, which is particularly important for the stripe patterns in the horse to zebra translation.

The network applies five convolutional layers with progressively increasing channel depth. LeakyReLU activations with a slope of 0.2 are used throughout. Instance normalisation is applied to all layers except the first, which is left unnormalised to preserve low-level colour and texture statistics. The final layer produces a patch-level map of real-versus-fake scores with no sigmoid activation, as the loss function operates directly on the raw outputs. Each discriminator has an effective receptive field of 70 by 70 pixels and contains approximately 2.8 million parameters.

The full system comprises two generators (one for each translation direction) and two discriminators (one for each domain), giving a total of approximately 28.4 million trainable parameters.

4.3 Proposed Modification: Perceptual Cycle Loss

The proposed variant of the architecture introduces a change only to the loss function used for cycle consistency training. The generator and discriminator network structures are identical to the baseline. The modification involves adding a frozen, pretrained VGG-16 network that serves as a fixed feature extractor. This network is never updated during training. Features are extracted from four intermediate layers of VGG-16, corresponding to progressively deeper levels of abstraction. The cycle consistency penalty is then computed in this feature space rather than directly on pixel values, as described in Section 5.

5. Loss Function

The total training objective combines three types of loss, each targeting a different aspect of the translation quality.

5.1 Adversarial Loss

The adversarial loss encourages the generator to produce images that are indistinguishable from real images in the target domain. This project uses the Least Squares GAN formulation (Mao et al., 2017) rather than the original binary cross-entropy loss. The least squares objective is more stable during training and tends to produce sharper results because it penalises samples that lie far from the decision boundary, rather than those that merely fool the discriminator with high confidence.

The discriminator is trained to output a value of 1 for real images and 0 for generated images. The generator is trained to make the discriminator assign a value of 1 to its outputs. An adversarial loss is applied independently for both translation directions.

5.2 Cycle Consistency Loss

The cycle consistency loss addresses the core challenge of unpaired training. Without a direct pixel-level supervision signal, the adversarial loss alone cannot constrain which horse image maps to which zebra image, leading to mode collapse. Cycle consistency enforces that if a horse image is translated to the zebra domain and then back to the horse domain, the result should closely match the original horse image, and vice versa.

In the baseline model, this constraint is enforced with an L1 penalty in pixel space. The L1 loss is weighted by a factor of 10, following the ablation study in Zhu et al. (2017). In the proposed modification, this pixel-space penalty is replaced by a perceptual loss. The reconstructed image and the original image are passed through a frozen VGG-16 network, and the difference between their feature maps is minimised at four intermediate layers (relu1_2, relu2_2, relu3_2, and relu4_2). The perceptual loss is applied with the same overall weight of 10, with equal contribution from each VGG layer. The rationale is that feature-space distances better capture perceptual and structural similarity than raw pixel differences, leading to sharper reconstructions.

5.3 Identity Loss

The identity loss is an auxiliary regularisation term that encourages the generator to act as a near-identity mapping when given an image that already belongs to the target domain. For example, passing a zebra image through the horse-to-zebra generator should return an image close to the original zebra. This helps preserve the colour characteristics of the input and prevents unnecessary changes to images that do not need to be translated. The identity loss is computed using an L1 penalty in pixel space in both the baseline and modified models, with a weight of 5 (equal to half the cycle consistency weight).

5.4 Summary of Loss Weights

The table below summarises the three loss terms and their weights in both the baseline and proposed modified models:

Loss Term	**Weight**	**Baseline**	**Modified**
Adversarial loss	1.0	L1 pixel (LSGAN)	L1 pixel (LSGAN)
Cycle consistency loss	10	L1 in pixel space	MSE in VGG-16 feature space

Loss Term	Weight	Baseline	Modified
Identity loss	5	L1 in pixel space	L1 in pixel space

6. Hyperparameters

All core hyperparameters were adopted directly from the original CycleGAN paper (Zhu et al., 2017) to ensure a faithful and reproducible baseline. Deviations were made only where hardware constraints required it, as noted in the table below.

Parameter	Value	Selection Rationale
Image size	256 x 256	Standard benchmark resolution for Horse2Zebra
Batch size	4	Constrained by available GPU memory (16 GB VRAM)
Learning rate	0.0002	Adopted directly from Zhu et al. (2017)
Optimizer	Adam	Standard choice for adversarial training
Adam beta 1	0.5	Recommended for GAN stability by Radford et al. (2016)
Adam beta 2	0.999	Default Adam value
Total epochs	200	Standard CycleGAN training schedule
LR decay start epoch	100	Linear decay from epoch 100 to 200, reaching zero
Residual blocks	9	Recommended by Zhu et al. for 256x256 inputs
Cycle loss weight	10	Selected via ablation study in Zhu et al. (2017)
Identity loss weight	5	Set to half the cycle loss weight, per Zhu et al.
Replay buffer size	50 images	Following recommendation of Shrivastava et al. (2017)
Mixed precision (AMP)	Enabled	Used for training efficiency on GPU
Data loading workers	8	Matched to the number of CPU cores available

The learning rate is held constant at 0.0002 for the first 100 epochs and then linearly decayed to zero over the remaining 100 epochs. This schedule allows the model to explore the parameter space freely in the early stages and then refine details as training progresses.

The replay buffer stores up to 50 previously generated images and is used to update the discriminator. At each discriminator update, either the current generated image or a randomly selected image from the buffer is used, each with 50 percent probability. This prevents the discriminator from adapting too rapidly to the current state of the generator, which improves training stability.

7. SOTA Comparison

7.1 Quantitative Comparison

The standard metric for evaluating image translation quality is the Frechet Inception Distance (FID). FID measures the statistical distance between the distribution of generated images and the distribution of real images, as estimated in the feature space of a pretrained Inception-v3 network. Lower FID values indicate better quality.

The table below reports published FID scores on the Horse2Zebra benchmark for several representative methods, along with the status of our own results:

Method	**FID H to Z (lower is better)**	**FID Z to H (lower is better)**	**Year**
CycleGAN (Zhu et al.)	77.2	138.1	2017
UNIT (Liu et al.)	96.1	145.3	2017
MUNIT (Huang et al.)	74.8	136.8	2018
CUT (Park et al.)	45.5	85.9	2020
Our Baseline (Epoch 150)	97.04	148.25	2026
Our Modified Model	Not available	Not available	2026

Quantitative evaluation of our baseline model was carried out on the standard Horse2Zebra test set (120 horse images in testA, 140 zebra images in testB) using the trained epoch 150 checkpoint. The FID for Horse to Zebra translation is 97.04 and for Zebra to Horse is 148.25. For reference, the original CycleGAN paper reports 77.2 and 138.1 respectively for a fully trained 200-epoch model. The gap of approximately 20 FID points on the Horse to Zebra direction is consistent with the fact that training was halted at epoch 150, halfway through the linear learning rate decay phase (epochs 100 to 200). The remaining 50 epochs of refinement are expected to bring the scores closer to the published values. The modified perceptual loss model was implemented but could not be trained or evaluated, as described in Section 8.

In addition to FID, cycle consistency was measured using the Structural Similarity Index (SSIM) between original images and their cycle-reconstructed counterparts. The Horse cycle SSIM is 0.8450 and the Zebra cycle SSIM is 0.8698, giving a mean SSIM of 0.8584. These scores indicate strong cycle consistency, confirming that the cycle consistency loss is functioning correctly and the model preserves structural content through the full round-trip translation.

7.2 Qualitative Comparison

Qualitative results for the baseline model are available from the training run, with sample outputs saved at every 10th epoch from epoch 10 to epoch 150. The following observations summarise the translation quality:

- Horse to Zebra: By epoch 80, the generator produces convincing stripe patterns that follow the body contours of the horse. By epoch 150, the texture is well-defined and perceptually plausible. Background regions are largely unaffected. The results are qualitatively comparable to the published examples in the original CycleGAN paper.
- Zebra to Horse: Stripe removal is partially successful, but the synthesised coat texture can appear inconsistent or overly smooth in some cases. This asymmetry between the two translation directions is a known limitation of the CycleGAN framework and is documented in the original paper. Adding stripes to a horse is an additive operation, whereas removing zebra stripes requires synthesising plausible smooth texture beneath a strong high-frequency pattern, which is inherently more difficult.
- Training stability: Generator loss decreased steadily from approximately 3.6 at epoch 71 to approximately 2.7 at epoch 150. Discriminator loss remained in the range of 0.17 to 0.54 throughout, indicating stable adversarial training without collapse or oscillation.
- Metric consistency: The directional asymmetry observed qualitatively is confirmed by both metrics. FID is better in the Horse to Zebra direction (97.04) than Zebra to Horse (148.25), while SSIM is slightly higher for the Zebra cycle (0.8698) than the Horse cycle (0.8450). The stronger SSIM for Zebra cycle despite the weaker FID for that direction reflects the different aspects each metric captures: FID measures the realism of the generated images, whereas SSIM measures how faithfully the original is reconstructed after the full translation round-trip.

7.3 Positioning Against the State of the Art

Our baseline achieves an FID of 97.04 on Horse to Zebra at epoch 150, compared to the published CycleGAN score of 77.2 at epoch 200. This places our partially trained model between the original CycleGAN and UNIT (96.1) on this direction, and within a plausible range of the published CycleGAN result given the incomplete training. The current strongest published result on this benchmark is CUT (Park et al., 2020), which achieves an FID of 45.5 using a contrastive learning objective rather than cycle consistency. Closing the gap with CUT is outside the scope of this project, but the proposed perceptual loss modification is intended as a step toward improving over the original CycleGAN baseline by reducing reconstruction blurriness. This hypothesis remains to be validated experimentally once GPU resources are available.

8. Remaining Work and Limitations

Due to loss of access to the university GPU cluster before the project deadline, several planned components could not be executed. All code is fully implemented, reviewed, and ready to run. The outstanding items are purely a matter of compute access. The table below summarises the status of each component:

Component	Status	Reason if Incomplete
Baseline model code	Complete	—
Modified perceptual loss model code	Complete	—
Baseline training, epochs 1 to 150	Complete	—
Inference web application (Gradio)	Complete	—
Baseline training, epochs 151 to 200	Incomplete	No GPU access
Modified model training	Not started	No GPU access
Quantitative evaluation of baseline (FID, SSIM)	Complete	—
Quantitative evaluation of modified model	Not run	No GPU access + no modified checkpoint
VGG layer ablation study	Not run	No GPU access and no modified checkpoints
Loss logs for epochs 1 to 70	Missing	Logs not retained from earlier session

8.1 Impact of Missing Components

The most significant missing component is the trained modified model. The central research contribution of this project is the hypothesis that replacing pixel-level cycle consistency with a perceptual feature-space loss improves translation quality. Without experimental results from the modified model, this hypothesis cannot be validated or refuted. The implementation is correct and architecturally sound, but the claim remains theoretical at this stage.

The incomplete final 50 epochs of baseline training are of moderate impact. The achieved FID of 97.04 on Horse to Zebra is approximately 20 points above the published 200-epoch result of 77.2. This gap is directly attributable to the incomplete learning rate decay schedule. The qualitative outputs at epoch 150 are strong and the SSIM cycle consistency of 0.8584 confirms the model is functioning correctly.

Quantitative evaluation of the baseline model has been completed. FID scores of 97.04 (Horse to Zebra) and 148.25 (Zebra to Horse) and SSIM cycle consistency scores of 0.8450 and 0.8698 respectively are available and reported in Section 7.

8.2 Planned Future Work

- Complete baseline training to epoch 200 and compute FID and SSIM scores on the full test set.
- Train the modified perceptual loss model for 200 epochs and run a full quantitative and qualitative comparison against the baseline.
- Execute the VGG layer ablation study, evaluating the modified model with cycle consistency measured at each of the four VGG feature levels independently to determine which level of abstraction is most effective.
- Investigate methods to reduce the Zebra to Horse quality gap, such as applying asymmetric loss weights that impose stronger cycle consistency on the harder direction.
- Extend the comparison to include CUT (Park et al., 2020) as a stronger modern baseline.

References

[1] Zhu, J.-Y., Park, T., Isola, P., and Efros, A. A. (2017). Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks. IEEE International Conference on Computer Vision (ICCV).

[2] Johnson, J., Alahi, A., and Fei-Fei, L. (2016). Perceptual Losses for Real-Time Style Transfer and Super-Resolution. European Conference on Computer Vision (ECCV).

[3] Mao, X., Li, Q., Xie, H., Lau, R. Y. K., Wang, Z., and Smolley, S. P. (2017). Least Squares Generative Adversarial Networks. IEEE International Conference on Computer Vision (ICCV).

[4] Isola, P., Zhu, J.-Y., Zhou, T., and Efros, A. A. (2017). Image-to-Image Translation with Conditional Adversarial Networks. IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

[5] Simonyan, K. and Zisserman, A. (2014). Very Deep Convolutional Networks for Large-Scale Image Recognition. International Conference on Learning Representations (ICLR 2015).

[6] Park, T., Efros, A. A., Zhang, R., and Zhu, J.-Y. (2020). Contrastive Learning for Unpaired Image-to-Image Translation. European Conference on Computer Vision (ECCV).

[7] Huang, X., Liu, M.-Y., Belongie, S., and Kautz, J. (2018). Multimodal Unsupervised Image-to-Image Translation. Advances in Neural Information Processing Systems (NeurIPS).

[8] Liu, M.-Y., Breuel, T., and Kautz, J. (2017). Unsupervised Image-to-Image Translation Networks. Advances in Neural Information Processing Systems (NeurIPS).

[9] Shrivastava, A., Pfister, T., Tuzel, O., Susskind, J., Wang, W., and Webb, R. (2017). Learning from Simulated and Unsupervised Images through Adversarial Training. IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

[10] Radford, A., Metz, L., and Chintala, S. (2016). Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks. International Conference on Learning Representations (ICLR).